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Automatic Motor Imagery Classification by CNN-Transformer-LSTM Using Multi-Channel EEG Signals

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Abstract. The brain-computer interface (BCI) is a promising technology that could bring about a significant revolution in various fields, including healthcare and human enhancement. One commonly used BCI method in healthcare, particularly in rehabilitation, is the analysis of motor imagery (MI) through an electroencephalogram (EEG). Our study introduces a hybrid deep learning model called CNN-Transformer-LSTM, which utilizes multi-channel EEG signals to classify MI binary and multiclass automatically. Our experiments have shown that this proposed method is more effective than previous state-of-the-art studies at accurately classifying MI using multi-channel EEG signals.

1 Introduction

The Brain-Computer Interface (BCI) system establishes a direct connection between the human brain and the external environment, functioning as a real-time brain-machine interface. This interface facilitates interaction with external parameters such as position, orientation, and movement. By harnessing the user's brain activity signals as a medium for communication, the BCI system translates these signals into the necessary output. Consequently, users can operate external devices not controlled by peripheral nerves or muscles solely through their brain activity [7].

The most popular current non-invasive method for acquiring brainwaves for BCI is detection through EEG. The popularity of EEG is supported by the inexpensiveness of the equipment, reduced complications compared to invasive procedures, portability, easy process to set up and use, and the possibility to measure neural activity directly. Of course, many disadvantages remain such as getting a very rough picture of the brain functionality or low spatial resolution. EEG reading is easy to use and has a high potential to be applied by the majority of the population to use a variety of possibilities that BCI offers. Other non-invasive methods include functional magnetic resonance imaging (fMRI), magnetoencephalography (MEG), and near-infrared spectroscopy (NIRS), which can be used separately or combined [1, 8]. Current BCI systems and applications rely on several fundamental paradigms, including detecting the brain frequencies, event-related or evoked components, (Steady-State) Visual Evoked Potentials (SSVEPs, VEPs), and Motor Imagery (MI).

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In the realm of BCI systems, the amalgamation of neuroscience and artificial intelligence (AI) fosters remarkable advancements with tangible real-world implications. By harnessing the power of AI algorithms, BCI systems decipher the complex patterns of brain activity associated with MI, including communication and control for individuals with severe motor disabilities, robot rehabilitation, virtual reality interaction, and cognitive assessment. MI holds particular significance in BCI applications due to its ability to allow patients to control robotic limbs or devices through their thoughts alone. By detecting and analyzing EEG signals associated with MI, researchers can decipher the neural mechanisms underlying MI tasks, leading to advancements in both basic understanding and AI application in the practice of BCI technology.

During MI, individuals typically experience decreased EEG amplitude in the alpha and beta bands, a phenomenon known as Event-Related Desynchronization (ERD). This reduction in EEG power reflects cortical activation associated with the mental rehearsal of movement sequences. The ERD phenomenon is often observed in the time window starting a few hundred milliseconds before the onset of the imagined movement and continuing throughout the MI task.

The MI task involves mentally simulating the execution of a physical movement without actually performing it. In this task, individuals are typically instructed to vividly imagine themselves performing a specific movement, such as moving their hand or foot, without any actual muscle movement. This mental rehearsal of movement sequences activates similar neural pathways as the actual execution of the movement, leading to detectable changes in brain activity, particularly in regions of the brain associated with motor control. During rest or in the absence of motor imagery, there is often an increase in EEG amplitude in the alpha and beta bands, termed Event-Related Synchronization (ERS). This augmentation in EEG power suggests a state of cortical idling or inhibition, contrasting with the engaged state observed during MI [4, 10].

It is crucial to identify MI patterns in real-time during the patient's use of the rehabilitation robot to provide timely feedback and guidance to patients, aiding in their recovery process. In this study, we proposed and implemented a CNN-Transformer-LSTM model for automatic MI classification using multi-channel EEG signals. The main contributions include introducing a novel architecture of the CNN-Transformer-LSTM model for MI classification (binary classification distinguishing resting state and hand movement and multi-

class classification distinguishing resting state, left-hand movement, and right-hand movement) using multi-channel EEG signals, using data augmentation methods to improve classification and evaluating model performance against state-of-the-art methods. Our study highlights the potential of integrating cutting-edge technologies in neuroscience by combining AI methodology and BCI applications.

The remainder of the paper is structured as follows. State-of-the-art is presented in Section 2. In Section 3, Materials and Method, we introduce the dataset used, data preprocessing and augmentation, the architecture of the proposed model, and its training. Section 4 provides the results of using augmentation and classification methods on the MI dataset. Finally, the discussion and the conclusion are provided in sections 5 and 6.

2 State-of-the-art

In recent years, there's been a surge in research focusing on refining data augmentation techniques and harnessing deep learning for MI classification. In a recent study conducted by Yang et al. [17], the primary objective centered on identifying four distinct movements involving the left hand, right hand, tongue, and both feet. Their analysis of the gathered data revealed that tasks involving MI-induced ERD and ERS across the motor cortex's right and left hemispheres, mainly at C4 and C3 electrodes. The Cz electrode was affected primarily by MI of the feet and tongue. To augment the measured EEG signals, the researchers combined a conditional variational auto-encoder network (cVAE) with a generative adversarial network (GAN) for learning latent representations from EEG signals. They employed an encoder network to obtain the distributed samples of the EEG signals. They applied an adversarial learning mechanism to continuously optimize the parameters of the generator, discriminator, and classifier. For all subjects, the cross-validation metric accuracy was several percent higher when the generated data were included in the dataset.

Schirrmeister et al. [12] designed and trained deep ConvNets (deep learning with convolutional neural networks) to decode MI tasks (hand movement, feet movement, and rest) from the raw EEG without handcrafted features and highlighted the potential of deep ConvNets combined with advanced visualization techniques for EEG-based brain mapping. The first technique shows how much information about a specific feature is retained in the ConvNet in different layers. The second technique is to directly investigate the causal effects of the feature values on the ConvNet outputs. Their results show good performance as the widely used filter bank common spatial patterns (FBCSP) algorithm (mean decoding accuracies 82.1% FBCSP, 84.0% deep ConvNets).

Tang et al. [13] proposed a new method based on the deep convolutional neural network (CNN) to perform feature extraction and classification for single-trial MI EEG. Firstly, based on the spatio-temporal characteristics of EEG, a 5-layer CNN model (the first layer is the input layer; the second and third layers are hidden layers, which compose the feature extraction part, and the fourth and fifth layers compose the classification part) is built to classify MI tasks (left-hand and right-hand movement). The application of the CNN model resulted in an accuracy of 86.41±0.77% for the experimental dataset collected from two able-bodied subjects, surpassing other conventional classification methods: power with Support Vector Machine (SVM), Common Spatial Pattern (CSP) with SVM and AutoRegression (AR) with SVM.

Ak et al. proposed a novel method for classifying MI EEG signals for BCI. EEG signals are first converted into spectrogram images. In the deep learning (DL) stage, 400 images are obtained for each task (movement of the robot arm up, down, left, and right) as input to the GoogLeNet (a 22-layer deep CNN). After the DL was completed, the presented system was tested to imagine up, down, left, and right movements to control the movement of the robot arm. It is observed that the robot arm performs the desired movement with over 90% accuracy [2].

In another study, Tibrewal et al. [14] evaluated the effectiveness of DL models in capturing MI features, particularly in inefficient users who face difficulties in achieving the desired accuracy or performance when using BCI systems. EEG signals from 54 subjects who performed an MI task of left- or right-hand grasp were recorded to compare the performance of two classification approaches. In the machine learning (ML) approach, CSP was used for feature extraction, and then the linear discriminant analysis (LDA) model was employed to binary classify the MI task. In the DL approach, a CNN model was constructed on the raw EEG signals. The results showed that the CNN model improved the classification accuracy for all subjects within the range of 2.37 to 28.28%, but more importantly, this improvement was significantly larger for low performers.

Zhang et al. [19] experimented with various augmentation methods to improve the classification accuracy when using CNN for MI detection. Their experimentation utilized two distinct datasets for assessing classifiers and augmentation methodologies: the openly accessible BCI Competition IV dataset 1 and BCI Competition IV dataset 2b. Within the analysis, a four-second EEG signal, corresponding to the duration of MI, served as the focal point. Evaluation of augmentation methods was conducted using the Frechet Inception Distance (FID) metric. All methods except the Geometric Transformation (GT) improved classification accuracy. The best improvement (12.6%) was achieved using the convolutional GAN for BCI Competition IV dataset 2b.

3 Materials and Method

This section introduces the MI dataset and describes EEG signal preprocessing, the data augmentation techniques, the proposed model architecture, model training, and performance metrics used.

3.1 Dataset

The following MI dataset was gathered at the University of West Bohemia in the Czech Republic. Twenty-nine healthy individuals were involved in the study, including men aged 21-26 and women aged 18-23. These participants were assessed while performing tasks related to left and right-hand MI. The experiment protocol to acquire EEG data using the rehabilitation robot is briefly described further; for a more detailed description of the protocol and the experimental process, see [5].

The participant sits in a chair and holds the arm of the rehabilitation robot with his left or right hand. The actual experimental measurement is then performed by alternating resting and movement phases for the duration of the experiment (10 minutes). In the resting phase, the subject is prevented from moving the robot arm for 10 seconds. In the movement phase, the subject moves the rehabilitation robot's arm along a predefined trajectory for 20 seconds.

The final dataset, named "EEG motor imagery," consists of data from 29 healthy subjects. The data from Fz, Cz, Pz, F3, F4, P3, P4, C3, and C4 channels was collected. Anonymized data are freely available at https://zenodo.org/record/7893847 [6]. The source code is available at https://github.com/ducthien512/CNNTransformerLSTM.

3.2 Data Preprocessing

The raw signal from three EEG electrodes (Cz, C3, and C4) was selected, four-second epochs were created, the baseline correction was performed, the epochs were undersampled to 500 Hz and filtered with a band-pass filter with a finite impulse response and cutoff frequencies of 8-30Hz. Artifacts were rejected with the threshold of 100 microvolts, and the epochs were selected to balance the training classes. The measured voltages create the values of the feature vector. The data was randomly shuffled, and the resulting dataset was divided (80% of the data was used as a training set and 20% as a test set) [5].

3.3 Data Augmentation

To improve MI classification, the following data augmentation methods were used: noise injection (NI) [21], cVAE [9], and Conditional Wasserstein Generative Adversarial Network with Gradient Penalty (cWGAN-GP) [20]. All augmentation metrics, such as signal-to-noise ratio (SNR), root mean square error (RMSE), and cross-correlation (CC), were applied. During the augmentation process, the original dataset was doubled. It means that the augmented training set comprised one-half of the actual EEG collected data and half of the data generated by the augmentation method [6]. Figures 1 and 2 show data augmentation methods for binary and multiclass classification using two-dimensional t-SNE projection.

3.4 Model architecture

For MI classification using multi-channel EEG signals, we use the model we developed to improve sleep stage classification accuracy using the single-channel EEG signal on the ISRUC sleep dataset [11]. The CNN-Transformer-LSTM model includes three units: the Convolutional Neural Network (CNN), the Transformer Encoder, and the Long Short-Term Memory (LSTM) network. Finally, a Softmax activation function is used for accurate MI binary classification (resting and movement) and multiclass classification (resting state, left movement, and right movement). A schematic diagram of the proposed MI classification method is shown in Figure 3.

CNN. The CNN is deployed to extract salient time-invariant features from the EEG signal. This stage compresses the signal's information while retaining critical characteristics relevant to MI. In this study, the CNN model comprises three Convolution 2D layers (Filter, Kernel Size, Stride), three Batch Normalization and ReLU layers, and three Max Pooling 2D layers (Pool size). The last layer is a Dropout. Figure 4 presents the CNN architecture.

Transformer Encoder. The Transformer Encoder captures time-varying patterns within the extracted features. Its internal structure is divided into two sub-layers structured by Multi-Head Attention and Feed-forward Neural Network (FNN). Both sub-layers utilize residual connections around each of the two sub-layers, followed by layer normalization. That is, the output of each sub-layer is LayerNorm(x + Sublayer(x)), where Sublayer(x) is the function implemented by the sub-layer itself [16]. The Transformer Encoder architecture is shown in Figure 5.

LSTM. The LSTM network, a variant of RNNs, was proposed by Hochreiter and Schmidhuber in 1997 [3]. The LSTM network is composed of cells or blocks that are placed in order. Each cell transfers two states to the next cell, the cell state and the hidden state, and includes three kinds of gates, the forget, the input, and the output gate, respectively. There is an additional input node, typically computed with a "tanh" activation function. The input gate determines how much of the input node's value should be added to the current memory cell's internal state. The forget gate determines whether to keep the current value of the memory or flush it. The output gate determines whether the memory cell should influence the output at the current time step. Figure 6 illustrates the LSTM network.

3.5 Model Training

Five-fold and ten-fold cross-validation were performed; each classification metric's mean and standard deviation iteration were calculated. In addition to classification metrics, the training and inference (classification) time were measured. Training time was calculated as the total duration of five-fold and ten-fold cross-validation, and classification time was measured as the duration of class prediction of a single feature vector. Five-fold cross-validation was used to expedite the process, while ten-fold cross-validation was employed for more thorough evaluation despite its longer runtime. All classifiers were trained using GPU.

We employed the Adam optimizer during the training process and utilized "categorical cross-entropy" as the loss function. The initial learning rate was set at 0.001, and we incorporated the ReduceLROnPlateau class. This class dynamically reduces the learning rate if the designated metric fails to improve within the 'patience' limit. Early stopping was implemented to prevent overfitting when the validation loss remained unchanged for 20 epochs. The model was trained for 100 epochs and implemented with the TensorFlow framework. Traditional metrics (accuracy, precision, recall, F1-score, and area under the receiver operating characteristic curve (AUC)) were used to evaluate the classification results.

4 Results

4.1 Augmentation and classification performance

This section presents the results of data augmentation methods and MI binary and multiclass classification.

Data augmentation results for binary and multiclass classification are shown in Tables 1 and 2. The results indicate that the cVAE method achieved the highest SNR metrics for both binary and multiclass classification, with scores of 23.887 and 23.898, respectively. On the other hand, the lowest RMSE metrics are 1.344 and 1.364 using the cWGAN-GP method. Lastly, the CC metrics reached a value of 0.214 and 0.267 using the cVAE method.

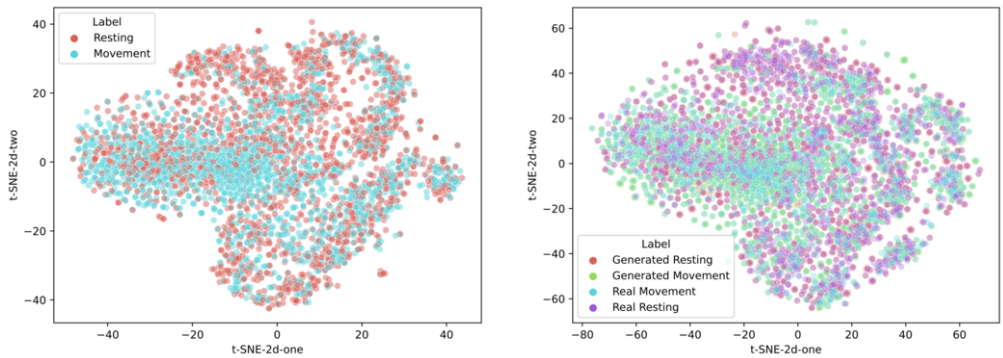
Table 1. Augmentation metrics for input data and binary classification.

Methods	SNR	RMSE	CC
NI	3.714	1.413	0.061
cVAE	**23.887**	1.420	**0.214**
cWGAN-GP	6.389	**1.344**	0.174

Table 2. Augmentation metrics for input data and multiclass classification.

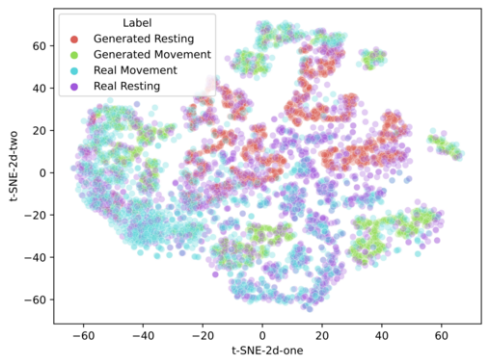
Methods	SNR	RMSE	CC
NI	3.739	1.412	0.072
cVAE	**23.898**	1.414	**0.267**
cWGAN-GP	6.819	**1.364**	0.227

Tables 3 and 4 present the results of MI binary and multiclass classification. Regarding binary classification, the highest performance is achieved in terms of accuracy (78.51±1.21), precision (78.58±1.19),

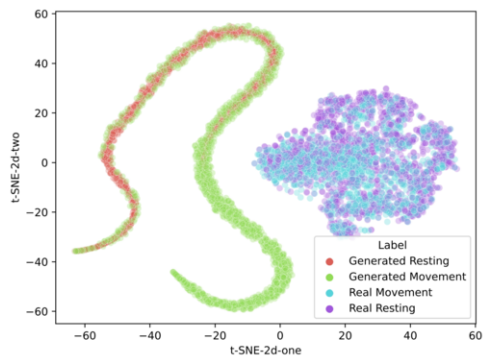


(a) No augmentation

(b) Augmentation with NI method

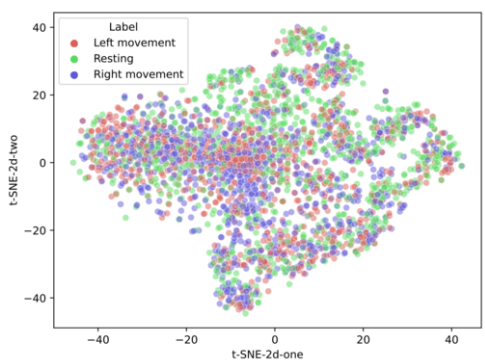


(c) Augmentation with cVAE method

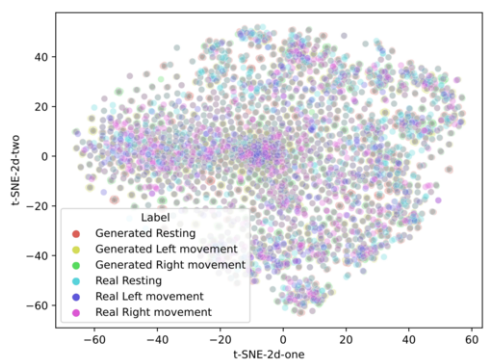


(d) Augmentation with cWGAN-GP method

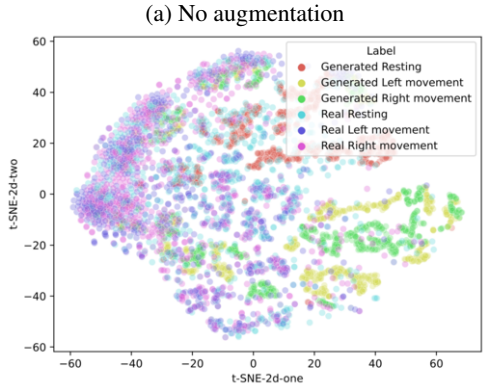
Figure 1. Data augmentation methods for binary classification on the EEG motor imagery dataset using two-dimensional t-SNE projection.



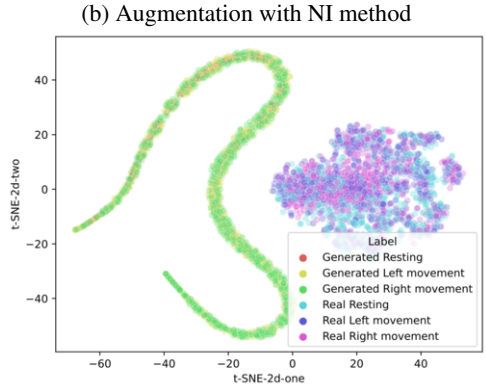
(a) No augmentation



(b) Augmentation with NI method



(c) Augmentation with cVAE method



(d) Augmentation with cWGAN-GP method

Figure 2. Data augmentation methods for multiclass classification on the EEG motor imagery dataset using two-dimensional t-SNE projection.

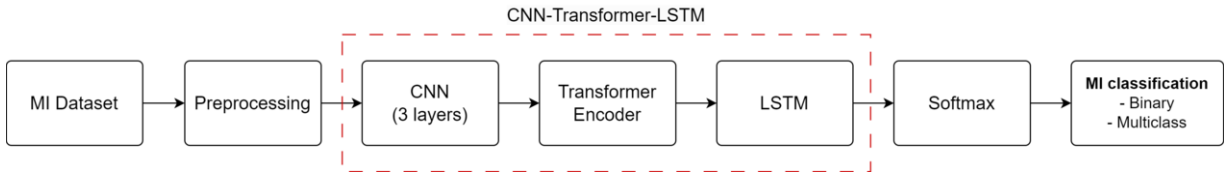


Figure 3. The architecture of the proposed method.

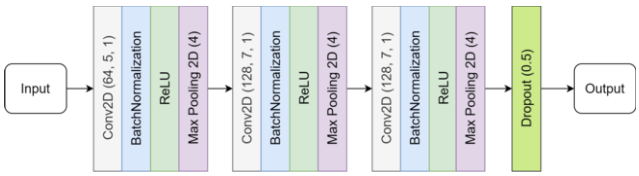


Figure 4. The architecture of the CNN.

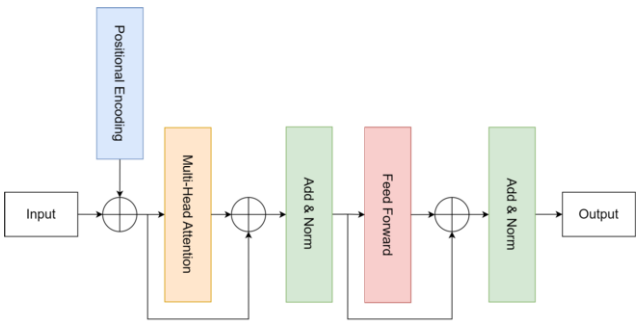


Figure 5. The architecture of the Transformer Encoder.

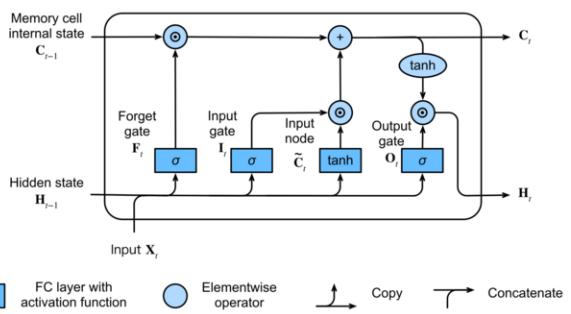


Figure 6. The architecture of the LSTM [18].

recall (78.52±1.21), F1-score (78.50±1.21), and AUC (0.785±0.01), using the CNN-Transformer-LSTM model without data augmentation method on ten-fold cross-validation for binary classification. For multiclass classification, the highest performance is achieved in terms of accuracy (61.95±1.99), precision (62.73±2.11), recall (61.84±2.07), F1-score (61.80±2.03), and AUC (0.715±0.02), using the CNN-Transformer-LSTM model without data augmentation method on ten-fold cross-validation. Figure 7 illustrates the confusion matrix for binary and multiclass classification. Figures 8 and 9 show the accuracy and loss of training and validation sets for binary and multiclass classification.

4.2 Performance Comparison

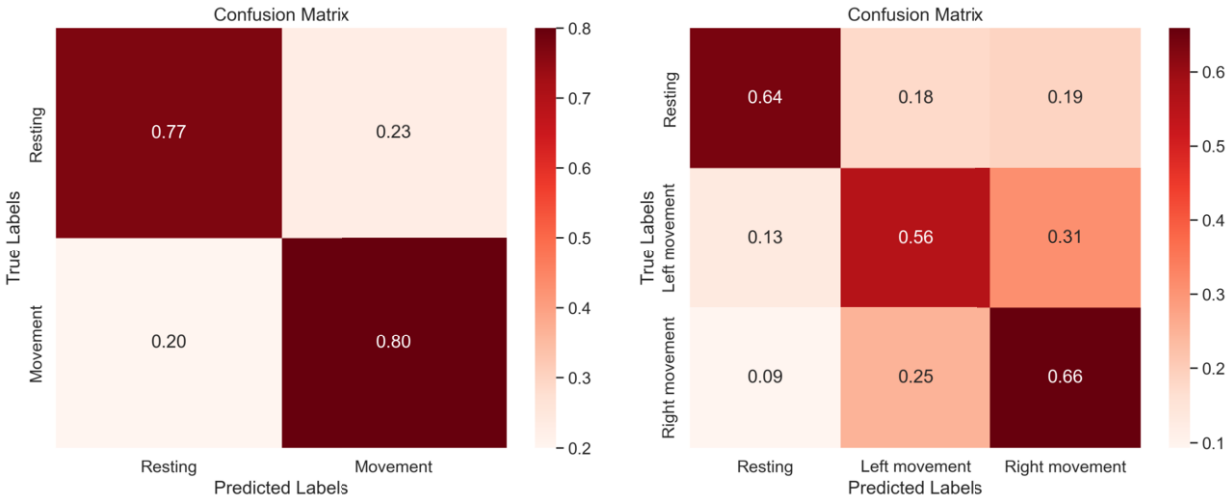
Based on the performance comparison outcomes of the MI binary and multiclass classification on the EEG MI dataset, as shown in Tables 5 and 6, the proposed model exhibits a notable improvement in classification accuracy. Specifically, it yields 2-3 percentage points higher classification results than the CNN [6] and EEG-ITNet [15] models, demonstrating its efficacy in handling the complexities inherent in MI data classification tasks. This enhancement underscores the effectiveness of the CNN-Transformer-LSTM model over existing state-of-the-art approaches.

5 Discussion

The CNN-Transformer-LSTM model achieved the highest results for binary and multiclass classification, with accuracies of 78.51±1.21% and 61.95±1.99%, respectively, without using augmentation. The proposed method achieves the highest performance metrics for binary and multiclass classification when comparing the results with state-of-the-art studies.

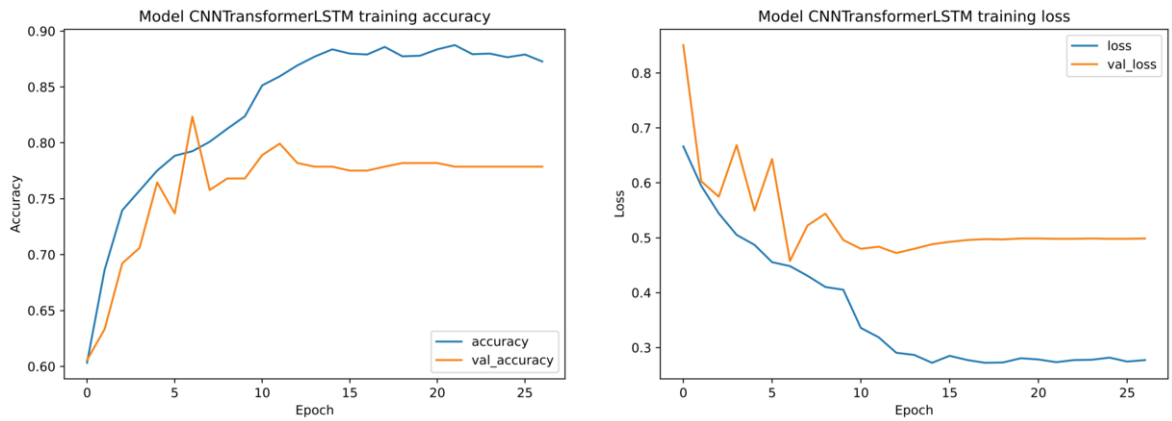
It is worth noting that the multiclass classification results were considerably lower than the binary classification results by more than 16 percentage points. This stark contrast underscores the inherent complexity of tackling multiclass classification tasks, particularly in discerning the intricate nuances of various hand movements. One explanation for this notable discrepancy lies in the inherent challenges associated with accurately distinguishing between left-hand and right-hand movements amidst a broader array of potential gestures and actions in the multiclass classification paradigm. The task gets tougher due to the increased complexity and detail, so multiclass classification requires a more complex and flexible classification model to extract features and achieve the best results.

Regarding the classification results when using three data augmentation methods (NI, cVAE, cWGAN-GP), the results from Tables 3 and 4 show that the NI method gives the best results among all three methods and the cVAE method gives the lowest results. Among them, the training process of CWGAN-GP consumes more time and energy than the remaining data augmentation methods. The synthetic samples produced by cWGAN-CP might not be as realistic or similar to the real data as intended. This could lead to a lack of overlap in the t-SNE plot if the synthetic data has different underlying structures or features compared to the original data, as shown in Figures 1 and 2. So although all the above approaches can generate the required data, the quality and diversity of the generated data differ from those of real data [20]. The classification results also show that the impact of augmentation methods is not significant. One possible explanation could be the inherent complexity and variability of MI data. Unlike other types of data, MI data often encapsulates subtle nuances and individualized patterns that are challenging to replicate accurately through augmentation methods. Moreover, the efficacy of data augmentation techniques heavily depends on the specific characteristics of the dataset and the chosen augmentation strategy. In the case of MI, the effectiveness of augmentation methods may be hindered by the limited diversity of available samples or the presence of inherent biases within the dataset.



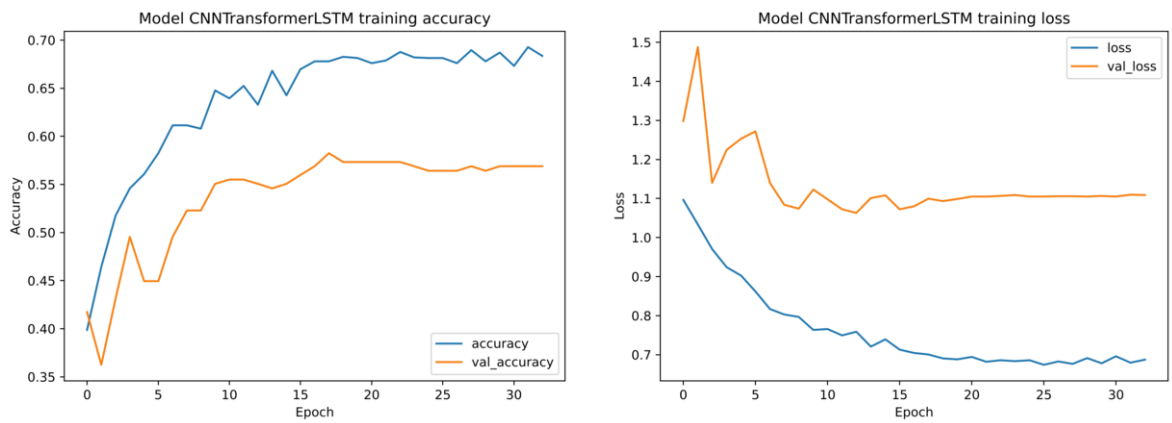
(a) (b)

Figure 7. The confusion matrix for (a) binary classification and (b) multiclass classification.



(a) (b)

Figure 8. The accuracy (a) and loss (b) for binary classification.



(a) (b)

Figure 9. The accuracy (a) and loss (b) for multiclass classification.

Table 3. Classification results for binary classification.

Fold	Methods	Accuracy	Precision	Recall	F1-score	AUC
5	CNN-Transformer-LSTM	78.40±0.66	78.72±0.46	78.43±0.65	78.35±0.70	0.784±0.01
	NI CNN-Transformer-LSTM	77.62±0.31	77.90±0.35	77.65±0.30	77.58±0.33	0.776±0.00
	cVAE CNN-Transformer-LSTM	77.37±0.52	77.61±0.52	77.40±0.52	77.33±0.54	0.774±0.01
	cWGAN-GP CNN-Transformer-LSTM	77.51±1.29	77.84±1.41	77.55±1.30	77.46±1.28	0.775±0.01
10	**CNN-Transformer-LSTM**	**78.51±1.21**	**78.58±1.19**	**78.52±1.21**	**78.50±1.21**	**0.785±0.01**
	NI CNN-Transformer-LSTM	78.31±1.11	78.40±1.11	78.32±1.11	78.30±1.12	0.783±0.01
	cVAE CNN-Transformer-LSTM	77.30±0.94	77.45±0.86	77.32±0.93	77.28±0.96	0.773±0.01
	cWGAN-GP CNN-Transformer-LSTM	77.46±1.13	77.88±1.02	77.50±1.13	77.38±1.16	0.775±0.01

Table 4. Classification results for multiclass classification.

Fold	Methods	Accuracy	Precision	Recall	F1-score	AUC
5	CNN-Transformer-LSTM	61.43±0.51	62.84±1.24	61.42±0.48	61.38±0.50	0.712±0.00
	NI CNN-Transformer-LSTM	60.40±1.04	60.46±1.11	60.07±1.09	60.10±1.09	0.702±0.01
	cVAE CNN-Transformer-LSTM	59.85±1.42	60.86±1.18	59.80±1.37	59.48±1.50	0.700±0.01
	cWGAN-GP CNN-Transformer-LSTM	59.89±2.23	61.06±2.10	59.75±2.18	59.47±2.27	0.700±0.02
10	**CNN-Transformer-LSTM**	**61.95±1.99**	**62.73±2.11**	**61.84±2.07**	**61.80±2.03**	**0.715±0.02**
	NI CNN-Transformer-LSTM	60.83±0.98	60.76±1.01	60.51±0.95	60.50±0.95	0.705±0.01
	cVAE CNN-Transformer-LSTM	60.55±1.41	61.66±1.62	60.49±1.46	60.30±1.53	0.705±0.01
	cWGAN-GP CNN-Transformer-LSTM	61.23±1.18	61.84±1.12	61.06±1.14	61.00±1.14	0.709±0.01

Table 5. Performance comparison with existing methods for binary classification on the EEG MI dataset.

Study	Methods	Accuracy	Precision	Recall	F1-score	AUC
[15]	EEG-ITNet	75.45±1.43	76.43±0.96	75.50±1.40	75.23±1.58	0.755±0.01
[6]	CNN	76.00±0.80	76.73±0.75	76.05±0.79	75.86±0.90	-
	Proposed method	**78.51±1.21**	**78.58±1.19**	**78.52±1.21**	**78.50±1.21**	**0.785±0.01**

Table 6. Performance comparison with existing methods for multiclass classification on the EEG MI dataset.

Study	Methods	Accuracy	Precision	Recall	F1-score	AUC
[6]	CNN	58.57±1.45	60.32±1.48	58.43±1.48	57.99±1.72	-
	Proposed method	**61.95±1.99**	**62.73±2.11**	**61.84±2.07**	**61.80±2.03**	**0.715±0.02**

To improve the classification performance, we propose the following measures: Increasing the amount of original data: it allows models to capture a broader range of patterns and variations present in the data, which can lead to better generalization; Using intra-models for each participant: it leverages the individual variability in MI data, potentially capturing participant-specific patterns more accurately. By treating each participant separately, the model can better adapt to the unique characteristics of their data. The effectiveness of these proposals would depend on the specific characteristics of the MI dataset, the goals of the classification task, and the available resources for data collection and model training.

Overall, the CNN-Transformer-LSTM model demonstrates improved classification results even without data augmentation. This emphasizes the significance of designing intricate architectures that align with the specific characteristics of the task and dataset. While data augmentation remains a valuable tool in many ML/DL applications, its necessity may vary depending on the inherent capabilities and architecture of the models employed.

6 Conclusion

This study presents the outcomes of augmentation and classification techniques using a CNN-Transformer-LSTM model when it is applied to a dataset comprising data from MI experiments. The binary (resting state and hand movement) and multiclass classification (resting state, left-hand movement, and right-hand movement) were performed. These experiments validate whether the MI concept holds promise for effective utilization in real BCI-controlled and robot-assisted motor rehabilitation. The proposed model shows promise for MI classification using EEG signals and highlights its potential to advance research in BCI applications and related fields, potentially alleviating the human workload.

Future work includes expanding the existing dataset or using other similar publicly available datasets, using intra-subject models, applying other DL/ML models (such as CNN-LSTM, CNN-Transfomer, etc.), or applying Transfer Learning to improve the accuracy of distinguishing between left-hand and right-hand movements. Moreover, combining EEG signals with EMG can improve the MI classification performance of the proposed method.

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